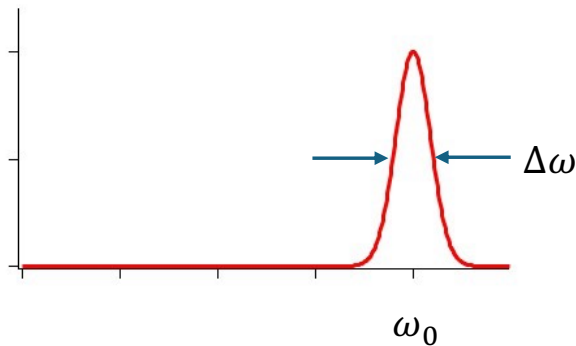


Atomic physics

Resonance

A resonance experiment is a method of observing changes in a system's response by varying the input frequency.



The ratio of the resonance frequency ω_0 to the width of resonance $\Delta\omega$ is called the “resonance Q value.”

$$Q = \frac{\omega_0}{\Delta\omega}$$

For precision measurements, it is important to build a system with a high Q value.

A simple example of resonance:
Forced oscillation of a damped harmonic oscillator

$$m \frac{d^2 x}{dt^2} = -m\gamma \frac{dx}{dt} - m\omega_0^2 x + f e^{i\omega t}$$

$$x(t) = x_0 e^{i\omega t}$$

$$m(\omega_0^2 - \omega^2 + i\gamma\omega)x_0 = f$$

$$\therefore x_0 = \frac{f}{m(\omega_0^2 - \omega^2 + i\gamma\omega)}$$

$$= \frac{f}{m(\omega_0 + \omega)(\omega_0 - \omega) + i\gamma\omega}$$

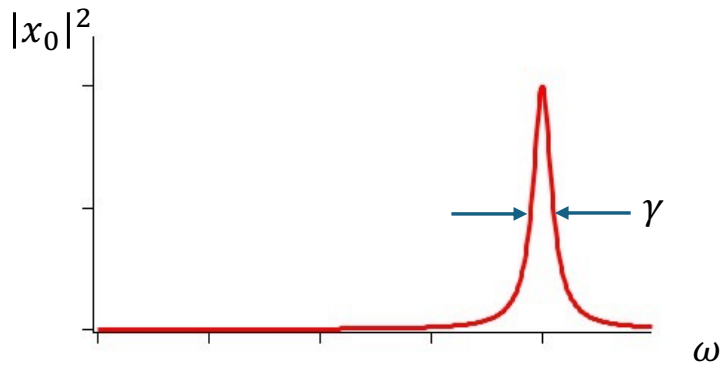
When $\omega \sim \omega_0$, we can substitute

$$(\omega_0 + \omega) \rightarrow 2\omega_0, \quad i\gamma\omega \rightarrow i\gamma\omega_0$$

$$x_0 \sim \frac{f}{2m\omega_0} \frac{1}{(\omega_0 - \omega) + i\frac{\gamma}{2}}$$

$$\therefore |x_0|^2 \propto \frac{1}{(\omega_0 - \omega)^2 + \frac{\gamma^2}{4}}$$

The value $(|x_0|^2)$ peaks at $\omega = \omega_0$, and it becomes half at $\omega = \omega_0 \pm \frac{\gamma}{2}$.



$$\therefore Q = \frac{\omega_0}{\gamma}$$

The Q value is also closely related to the system's behavior when it is not forced to oscillate.

When there is no external force,

$$\frac{d^2x}{dt^2} + \gamma \frac{dx}{dt} + \omega_0^2 x = 0$$

By writing $x(t) = x_0 e^{i\omega t}$

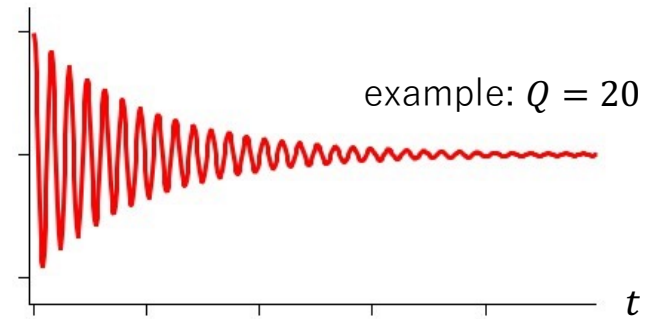
$$\omega = \frac{i\gamma \pm \sqrt{4\omega_0^2 - \gamma^2}}{2}$$

Assuming $\omega_0 \gg \gamma$

$$\omega \sim i\frac{\gamma}{2} \pm \omega_0$$

The solution becomes

$$x(t) = e^{-\frac{\gamma}{2}t} \cos \omega_0 t$$



Let's calculate the number of oscillations required for the amplitude to become $1/e$ of the beginning

$$\frac{2}{T} = \frac{2\omega_0}{\gamma 2\pi} = \frac{Q}{\pi}$$

We can see that the Q value corresponds to the number of times the oscillator oscillates before it decays.

Example of high Q oscillator: atom

Spontaneous emission rate Γ

$$\Gamma = \frac{1}{3\pi\epsilon_0} \frac{\omega_0^3}{\hbar c^3} d^2 \cong \frac{e^2}{3\pi\epsilon_0} \frac{\omega_0^3}{\hbar c^3} a_0^2 = \frac{e^2}{3\pi\epsilon_0} \frac{1}{\hbar a_0} \left(\frac{\omega_0}{c} a_0\right)^3$$

$$\therefore \frac{\Gamma}{\omega_0} \cong \left(\frac{\omega_0}{c} a_0\right)^3$$

$$\text{From } \hbar\omega_0 \cong \frac{e^2}{4\pi\epsilon_0} \frac{1}{a_0}, \frac{\omega_0}{c} a_0 \cong \frac{e^2}{4\pi\epsilon_0 \hbar c} = \alpha \cong \frac{1}{137}$$

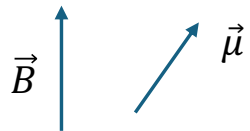
$$Q = \frac{\omega_0}{\Gamma} \propto \left(\frac{1}{\alpha}\right)^3 \sim 10^6$$

Precession of a Spin in a Magnetic Field

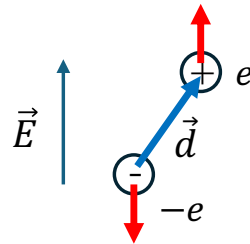
A magnetic moment in a magnetic field experiences a torque.

$$\vec{\tau} = \vec{\mu} \times \vec{B}$$

magnetic moment
in a magnetic field



electric dipole moment
in an electric field



$$\vec{\tau} = \vec{\mu} \times \vec{B}$$

$$\begin{aligned}\vec{\tau} &= \frac{\vec{r}}{2} \times (e\vec{E}) + \left(-\frac{\vec{r}}{2}\right) \times (-e\vec{E}) \\ &= (e\vec{r}) \times \vec{E} = \vec{d} \times \vec{E}\end{aligned}$$

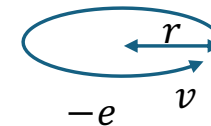
magnetic moment is proportional to the angular momentum

magnetic moment = current $\times$ area

$$\mu \sim \frac{-ev}{2\pi r} \pi r^2 = -\frac{erv}{2} = -\frac{ermv}{2m} = -\frac{e}{2m} l_z$$

in general, we introduce g-factor

$$\mu_z = -g \frac{e}{2m} l_z$$

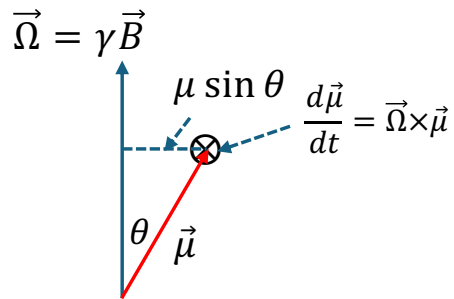


derivative of angular momentum is torque

$$\begin{aligned}\frac{d\vec{\mu}}{dt} &= -g \frac{e}{2m} \frac{d\vec{l}}{dt} = -g \frac{e}{2m} \vec{\tau} = -g \frac{e}{2m} \vec{\mu} \times \vec{B} \\ \therefore \frac{d\vec{\mu}}{dt} &= g \frac{e}{2m} \vec{B} \times \vec{\mu}\end{aligned}$$

using "gyromagnetic ratio" $\gamma = g \frac{e}{2m}$

$$\therefore \frac{d\vec{\mu}}{dt} = \gamma \vec{B} \times \vec{\mu}$$



The magnetic moment $\vec{\mu}$ rotates around $\vec{\Omega}$ where $\vec{\Omega} = \gamma \vec{B}$ is the angular velocity vector

When the magnetic field is constant and points towards z, (i.e. $\vec{B}_0 = \begin{pmatrix} 0 \\ 0 \\ B_0 \end{pmatrix}$), the time evolution of the magnetic moment is

$$\begin{pmatrix} \mu_x \\ \mu_y \\ \mu_z \end{pmatrix} = \mu \begin{pmatrix} \sin \theta_0 \cos(\gamma B_0 t + \phi_0) \\ \sin \theta_0 \sin(\gamma B_0 t + \phi_0) \\ \cos \theta_0 \end{pmatrix}$$

Next, let's add an oscillating magnetic field to this system. Suppose that the oscillating magnetic field rotates within the xy plane, and that its angular velocity matches the angular velocity of the precession.

$$\begin{aligned} \frac{d\vec{\mu}}{dt} &= \gamma \vec{B} \times \vec{\mu} \\ \vec{B} &= \begin{pmatrix} B_1 \cos \omega t \\ B_1 \sin \omega t \\ B_0 \end{pmatrix} \\ \omega &= \gamma B_0 \end{aligned}$$

Let's introduce rotating frame $\begin{pmatrix} x' \\ y' \\ z' \end{pmatrix}$ that rotates with angular velocity $\omega = \gamma B_0$

An object at rest in a rotating frame of reference is moving from the perspective of an inertial frame of reference. When $\left(\frac{d\vec{A}}{dt}\right)_{rot} = \vec{0}$, $\left(\frac{d\vec{A}}{dt}\right)_{inert} = \vec{\Omega} \times \vec{A}$.

Thus,

$$\left(\frac{d\vec{A}}{dt}\right)_{inert} = \left(\frac{d\vec{A}}{dt}\right)_{rot} + \vec{\Omega} \times \vec{A}$$

inertial frame

$$\frac{d\vec{\mu}}{dt} = \gamma \vec{B} \times \vec{\mu}$$

$$\vec{B} = \begin{pmatrix} B_1 \cos \omega t \\ B_1 \sin \omega t \\ B_0 \end{pmatrix}$$

$$\omega = \gamma B_0$$

rotating frame
 $\omega = \gamma B_0$

$$\frac{d\vec{\mu}'}{dt} = \gamma \vec{B}' \times \vec{\mu}' - \vec{\Omega} \times \vec{\mu}'$$

$$\vec{B}' = \begin{pmatrix} B_1 \\ 0 \\ B_0 \end{pmatrix} \leftarrow \text{oscillating magnetic field observed from rotating frame}$$

$$\vec{\Omega} = \begin{pmatrix} 0 \\ 0 \\ \omega \end{pmatrix}$$

Suppose the spin was pointing upwards when $t = 0$.

$$\begin{aligned} \frac{d\vec{\mu}'}{dt} &= \gamma \vec{B}' \times \vec{\mu}' - \vec{\Omega} \times \vec{\mu}' \\ &= \begin{pmatrix} \gamma B_1 \\ 0 \\ \gamma B_0 - \omega \end{pmatrix} \times \vec{\mu}' \\ &= \begin{pmatrix} \Omega_R \\ 0 \\ \omega_0 - \omega \end{pmatrix} \times \vec{\mu}' \end{aligned}$$

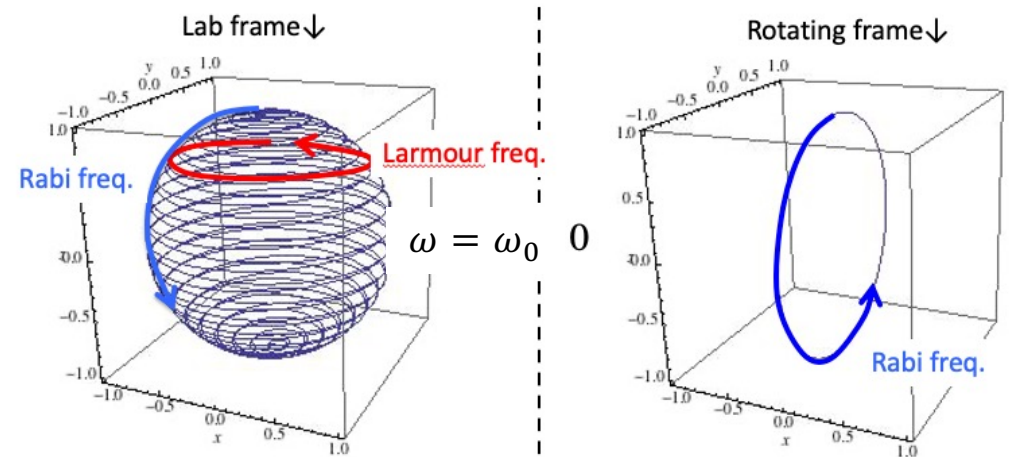
Here we defined

$\gamma B_0 = \omega_0$: Larmour frequency
 $\gamma B_1 = \Omega_R$: Rabi frequency

ω_0 is the frequency of the precession of the spin. If ω matches with ω_0 ,

$$\frac{d\vec{\mu}'}{dt} = \begin{pmatrix} \Omega_R \\ 0 \\ 0 \end{pmatrix} \times \vec{\mu}'$$

In the rotating frame, the spin rotates around x axis.

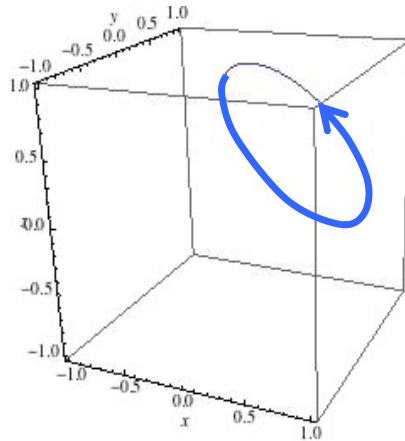


$$\vec{\mu}' = \mu \begin{pmatrix} 0 \\ -\sin \Omega_R t \\ \cos \Omega_R t \end{pmatrix}$$

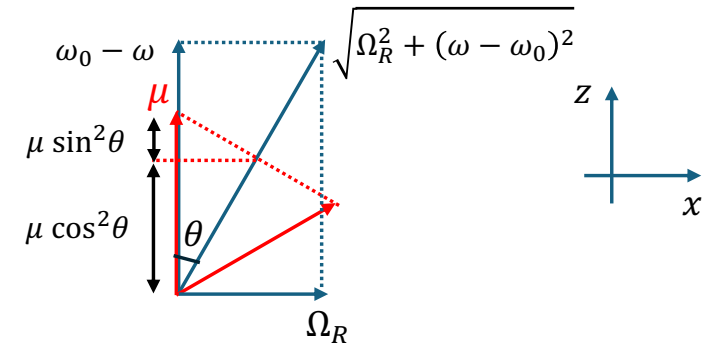
The spin precesses around the z-axis and undergoes Rabi oscillations around the x-axis.

Next, let's consider the case where the oscillating magnetic field does not satisfy the resonance condition ($\omega \neq \omega_0$). In this case, in the equations of motion for a rotational system, the spin's axis of rotation is tilted.

$$\frac{d\vec{\mu}'}{dt} = \begin{pmatrix} \Omega_R \\ 0 \\ \omega_0 - \omega \end{pmatrix} \times \vec{\mu}'$$



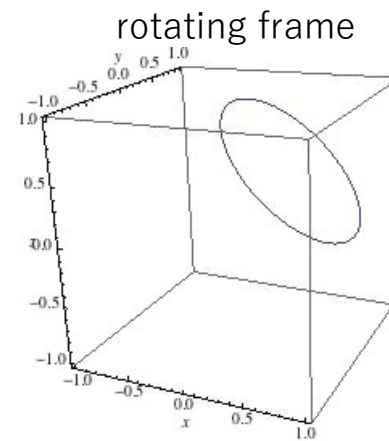
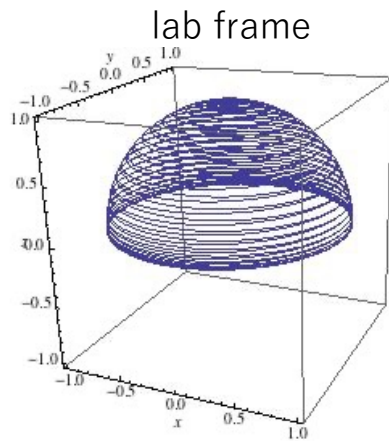
Suppose the spin was pointing upward at time $t = 0$



In the rotating frame, μ'_z oscillates between μ and $\mu(1 - 2\sin^2\theta)$ with angular frequency $\Omega'_R = \sqrt{\Omega_R^2 + (\omega - \omega_0)^2}$

$$\begin{aligned} \mu'_z &= \mu(1 - \sin^2\theta(1 - \cos(\Omega'_R t))) \\ &= \mu \left(1 - 2 \sin^2\theta \sin^2 \left(\frac{\Omega'_R t}{2} \right) \right) \\ &= \mu \left(1 - 2 \frac{\Omega_R^2}{\Omega'^2_R} \sin^2 \left(\frac{\Omega'_R t}{2} \right) \right) \end{aligned}$$

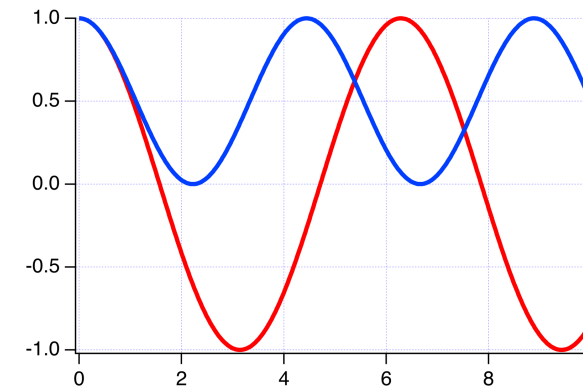
In an inertial frame, the spin precesses at the Larmor frequency and also flips up and down, but it returns to its original orientation before it can point completely downward.



Time evolution of z-component of the spin $\mu'_z(t)$ is

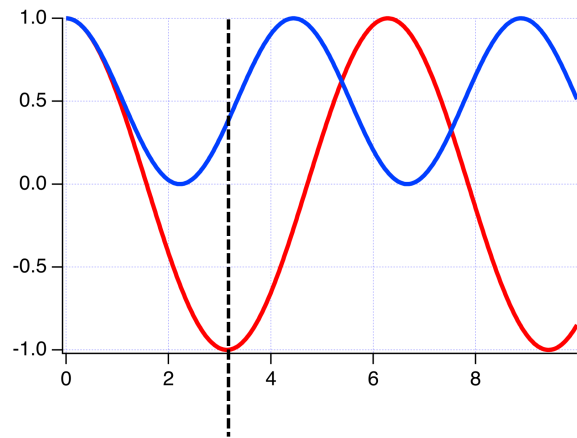
$$\frac{\mu'_z(t)}{\mu} = 1 - 2 \frac{\Omega_R^2}{\Omega_R^2 + (\omega - \omega_0)^2} \sin^2 \left(\frac{\sqrt{\Omega_R^2 + (\omega - \omega_0)^2}}{2} t \right)$$

となる。



red : on resonance ($\omega - \omega_0 = 0$)
blue : off resonance ($\omega - \omega_0 = \Omega_R$)

The spin inverts when $\omega = \omega_0$ and $t = \frac{\pi}{\Omega_R}$
→we call this " π pulse"

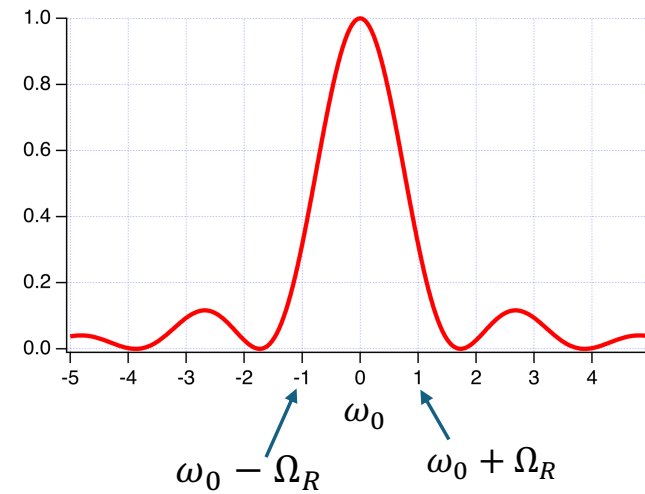


↑ Stop the oscillating magnetic field
at the timing of π pulse ($t = \frac{\pi}{\Omega_R}$)

Probability of spin flip:

$$P_{\downarrow}(\omega) = \frac{\Omega_R^2}{\Omega_R^2 + (\omega - \omega_0)^2} \sin^2 \left(\frac{\pi}{2\Omega_R} \sqrt{\Omega_R^2 + (\omega - \omega_0)^2} \right)$$

$$= \frac{1}{1 + \left(\frac{\omega - \omega_0}{\Omega_R} \right)^2} \sin^2 \left(\frac{\pi}{2} \sqrt{1 + \left(\frac{\omega - \omega_0}{\Omega_R} \right)^2} \right)$$



Resonance frequency is ω_0
Full Width of Half Max is $\sqrt{5}\Omega_R \sim 2.23\Omega_R$

MARCH 15, 1939

PHYSICAL REVIEW

VOLUME 55

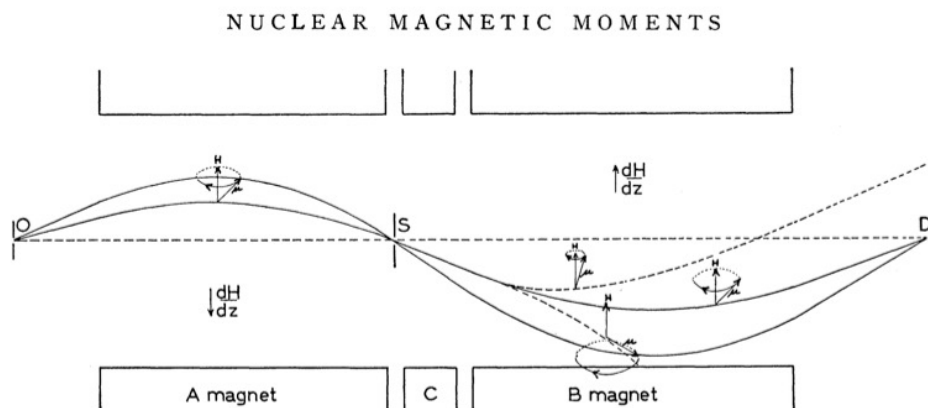
The Magnetic Moments of ${}^6_3\text{Li}$, ${}^7_3\text{Li}$ and ${}^{19}_9\text{F}$ *

I. I. RABI, S. MILLMAN, P. KUSCH, *Columbia University, New York, New York*

AND

J. R. ZACHARIAS, *Hunter College, New York, New York*

(Received January 20, 1939)



527

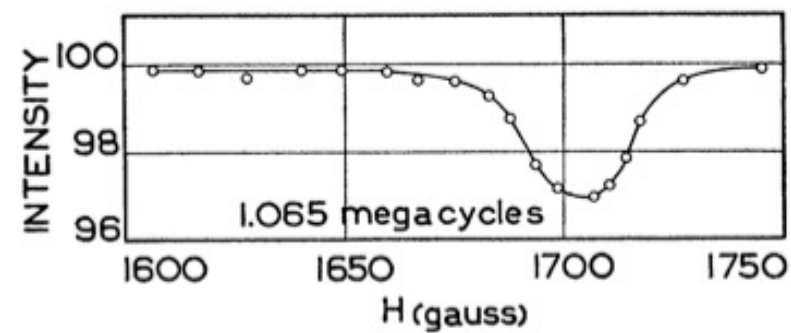


FIG. 3. Resonance curve of the Li^6 nucleus observed in LiCl .

g-factor of proton was measured in this way.

The biggest systematic error:
non-uniform magnetic field

OCTOBER 15, 1939

PHYSICAL REVIEW

VOLUME 56

The Magnetic Moments of the Proton and the Deuteron

The Radiofrequency Spectrum of H_2 in Various Magnetic Fields *

J. M. B. KELLOGG, I. I. RABI AND N. F. RAMSEY, JR.
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AND

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(Received July 31, 1939)

730

KELLOGG, RABI, RAMSEY AND ZACHARIAS

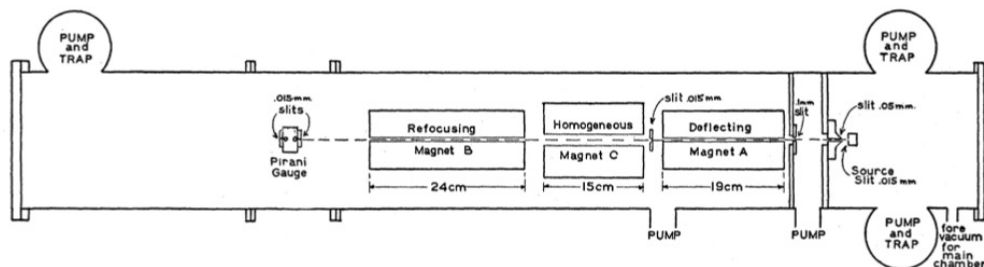
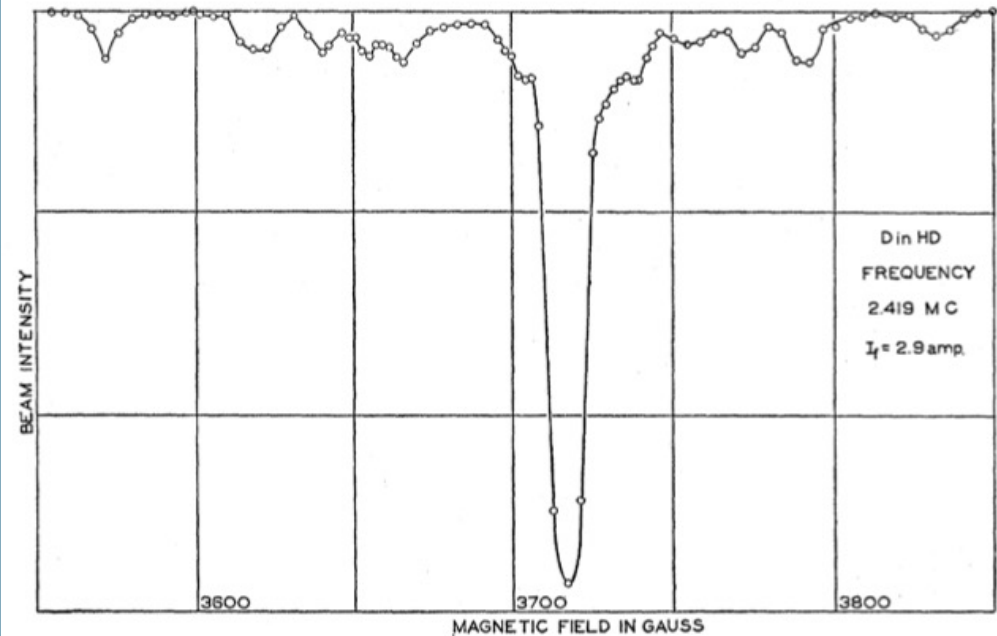


FIG. 1. Schematic diagram of apparatus. Longitudinal dimensions from source to detector are drawn to scale. Flattened copper tubes 13.5 cm long to carry radiofrequency current are not shown. They should appear between the pole pieces of magnet C.



stability of magnetic field $\sim 0.5\%$

In 1949, Ramsey proposed a new method.

N.F.Ramsey, Phys. Rev. 76, 996 (1949)

A New Molecular Beam Resonance Method*

NORMAN F. RAMSEY
Harvard University, Cambridge, Massachusetts
August 22, 1949

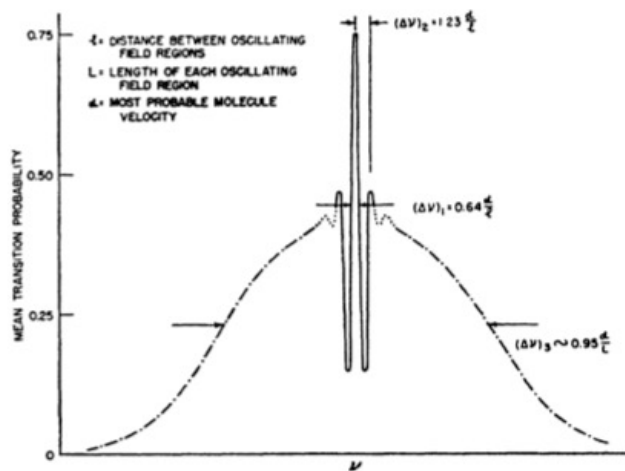
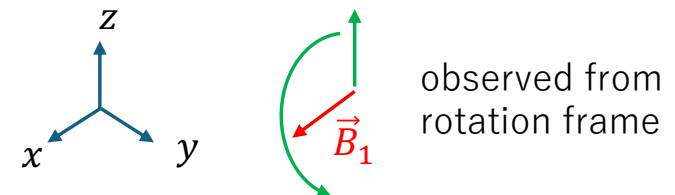
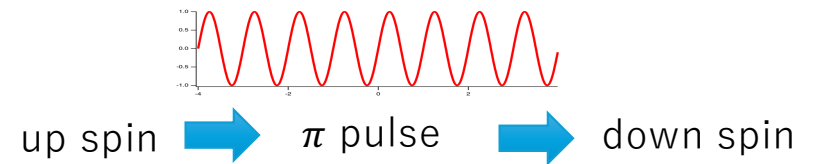


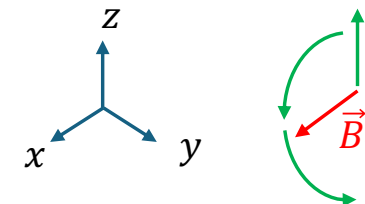
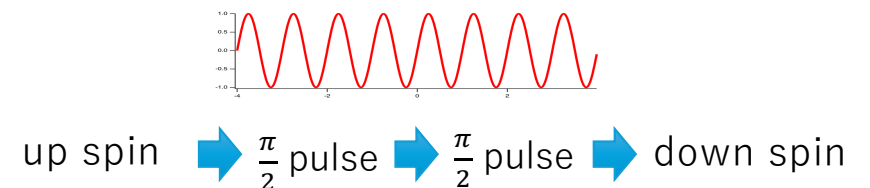
FIG. 2. Transition probability showing effect of finite length of oscillating field regions.

Divide π -pulse into two $\frac{\pi}{2}$ pulses and interfere those two

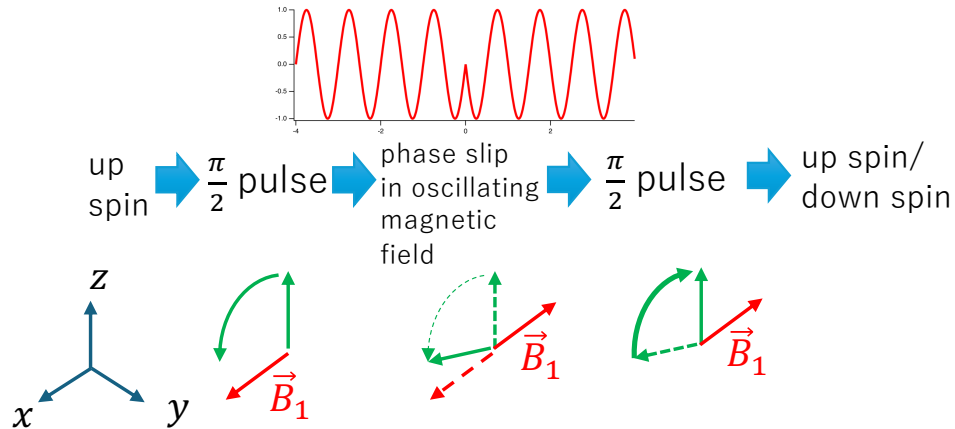
(1) π pulse



(2) two $\pi/2$ pulses



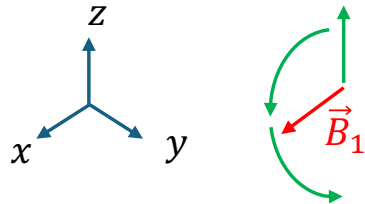
(3) $\frac{\pi}{2}$ pulse + phase slip in osc. field + $\pi/2$ pulse



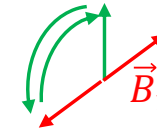
If the phase of the oscillating magnetic field is reversed partway through, the spin returns to its original state.

(4 : Ramsey method) $\pi/2$ pulse + osc. field off for time $T + \pi/2$ pulse

From the perspective of the rotating frame, if the phase of the oscillating magnetic field remains constant, the first $\pi/2$ pulse and the second $\pi/2$ pulse will add together to form a π pulse, causing the spin to flip.



If the phase of the oscillating magnetic field changes by π during time T , the situation described in (3) occurs, and the spin returns to the up direction.



→ The probability that the final spin points downward behaves as $(1 + \cos \phi)/2$ relative to the phase shift of the oscillating magnetic field, as seen from the rotating frame of reference.

If $\pi/2$ pulse exactly rotates the spin by $\pi/2$, the probability of finding the spin pointing downward is:

$$P_{\downarrow} \sim \frac{1 + \cos(\omega - \omega_0)T}{2} = \cos^2\left(\frac{(\omega - \omega_0)T}{2}\right)$$

In reality, we cannot neglect the frequency dependence of $\pi/2$ pulses. Thus, the probability of finding the spin pointing downward is affected in the following manner:

$$P_{\downarrow} = 4 \frac{\Omega_R^2}{\Omega_R'^2} \sin^2\left(\frac{\Omega_R'}{2}\tau\right) \times \left(\cos\left(\frac{(\omega - \omega_0)}{2}T\right) \cos\left(\frac{\Omega_R'}{2}\tau\right) - \frac{\omega - \omega_0}{\Omega_R'} \sin\left(\frac{(\omega - \omega_0)}{2}T\right) \sin\left(\frac{\Omega_R'}{2}\tau\right) \right)^2$$

Here, τ is the duration of $\pi/2$ pulse.

Experimental Advantages:

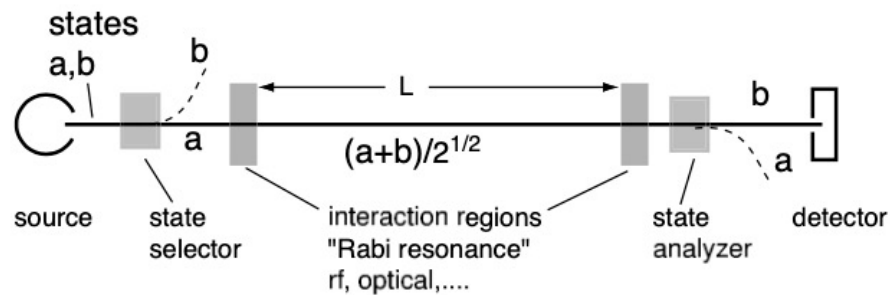
The accuracy of frequency measurements is proportional to the time T between two $\pi/2$ pulses.

- To improve accuracy, simply increase T .

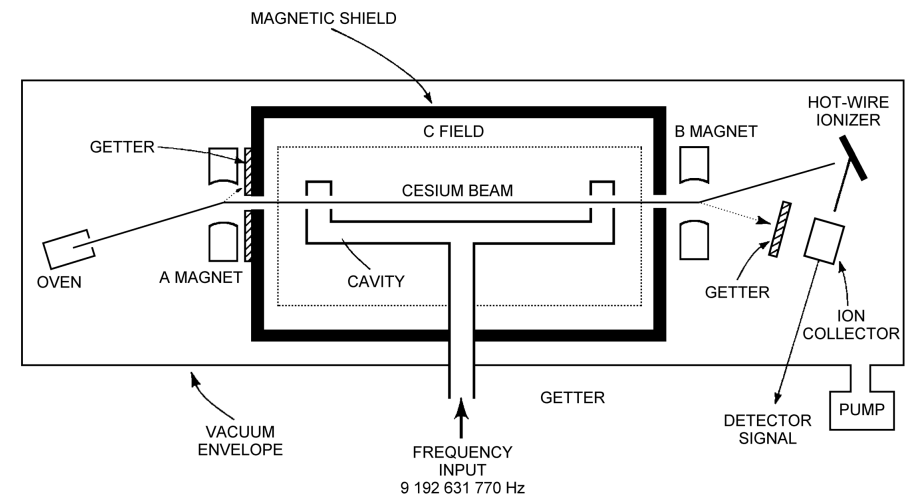
Furthermore,

- Even if the magnetic field in the measurement region fluctuates, the only factor that matters is the integrated phase.

→ It is acceptable even if the magnetic field is non-uniform.



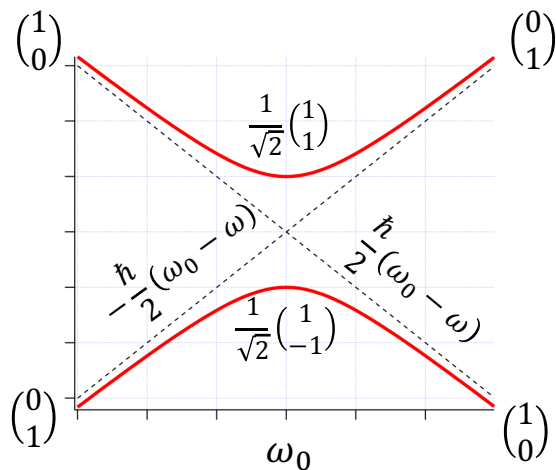
Early atomic clocks were also built using this method (First Generation)



Adiabatic Passage (Landau-Zener)

The value of eigen function $E(\omega)$ is as follows

$$E = \pm \frac{\hbar}{2} \sqrt{(\omega_0 - \omega)^2 + \Omega_R^2}$$



If the frequency is changed slowly, the system remains in its eigenstate.

→ This means we can flip the spin!

Probability of non-adiabatic process

$$P = e^{-2\pi\Gamma}$$

$$\Gamma = \frac{\frac{|V_{12}|^2}{\hbar^2}}{\frac{d\omega}{dt}}$$

$$\therefore P = e^{-\frac{2\pi(\frac{\Omega_R}{2})^2}{\frac{d\omega}{dt}}} = e^{-\frac{\Omega_R^2}{2\frac{d\omega}{dt}}}$$

Increasing the strength of the oscillating magnetic field (increasing Ω_R) allows for rapid spin reversal.

This method is called as "Adiabatic Rapid Passage"